

Chapter 2

Software Project Scheduling

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Objectives of Activity Planning:

Planning in software project management refers to the **process of defining, organizing, and documenting the various aspects of a software project** to ensure its successful execution.

It involves creating a **detailed roadmap** that outlines the project's scope, objectives, tasks, schedules, resources, risks, and quality standards.

Effective planning is crucial in software project management as it helps in delivering software products on time, within budget, and with the desired level of quality.

The objectives of activity planning are:

1. Feasibility Assessment:

- *Objective:* To determine whether a **proposed activity or project is viable and achievable**.
- *Explanation:*

Feasibility assessment as part of activity planning helps in evaluating the practicality and likelihood of success of the planned activity.

It involves considering factors such as technical, economic, legal, operational, scheduling, and resource feasibility.

The objective is to identify potential challenges and decide whether to proceed with the activity or make necessary adjustments.

2. Allocation of Resources:

- *Objective:* To **ensure** that the **right resources** (e.g., human resources, materials, budget) **are available and allocated effectively** to complete the planned activities.
- *Explanation:* Proper allocation of resources is a critical objective in activity planning. It ensures that there are **no resource shortages** or overages, leading to efficient resource utilization and preventing delays or disruptions in the project.

3. Estimation of Costs:

- *Objective:* To calculate the **anticipated costs associated with the planned activities**.
- *Explanation:* Cost estimation is essential to **budgeting and financial planning**. It helps in understanding the financial implications of the planned activities, including direct and indirect

costs. This objective ensures that the activity aligns with budget constraints and financial goals.

4. Project Coordination:

- *Objective:* To establish a clear framework for coordinating and managing various aspects of the project or activity.
- *Explanation:* Effective project coordination is crucial for ensuring that **all activities are carried out in a synchronized manner**.

It **involves assigning responsibilities, setting timelines, and facilitating communication** among team members and stakeholders.

The objective is to avoid conflicts, or misunderstandings that can hinder progress.

5. Personal Encouragement:

- *Objective:* To **motivate and empower individuals or teams** involved in the activity.
- *Explanation:* While it might not be a traditional objective of activity planning, personal encouragement is important for **maintaining motivation and morale among participants**.

It involves **recognizing and rewarding achievements, providing support and guidance,**

and fostering a positive work environment. Encouragement can **enhance productivity** and commitment to the project's success.

Project Schedule:

Can you **imagine starting a long car trip** to an **unfamiliar destination without a map** or navigation system?

You're pretty sure you have to make **some turns** here and there, but you have **no idea when or where**, or **how long it will take to get there**.

You may arrive eventually, but you **run the risk of getting lost, and feeling frustrated**, along the way.

Essentially, driving without any idea of how you're going to get there is the **same** as **working on a project without a schedule**.

No matter the size or scope of your project, the **schedule is a key part of project management**.

The schedule tells you **when each activity should be done**, **what** has **already been completed**, and the sequence in which things need to be finished.

Schedules also help you do the following:

- They provide a **basis** for you **to monitor and control project activities**.
- They help you determine **how best to allocate resources** so you can achieve the project goal.
- They help you **assess how time delays** will **impact** the project.
- You can figure out **where excess resources are available to allocate** to other projects.
- They provide a basis to **help** you **track project progress**.

To plan and schedule project activities and tasks the project manager needs to take the next four steps:

- Set up activities.
- Define relationships between activities.
- Estimate resources required for performing activities.
- Estimate durations for activities.

Work Breakdown Structure:

A Work Breakdown Structure (WBS) is a **hierarchical and systematic decomposition of a project into smaller, more manageable components, tasks, or work packages.**

The primary purpose of a WBS is to **help project managers and teams organize, define, and understand the scope of a project.**

A WBS is organized hierarchically, with the **highest-level component representing the entire project and subsequent levels breaking it down into increasingly detailed subcomponents.** This hierarchical structure forms a tree-like diagram.

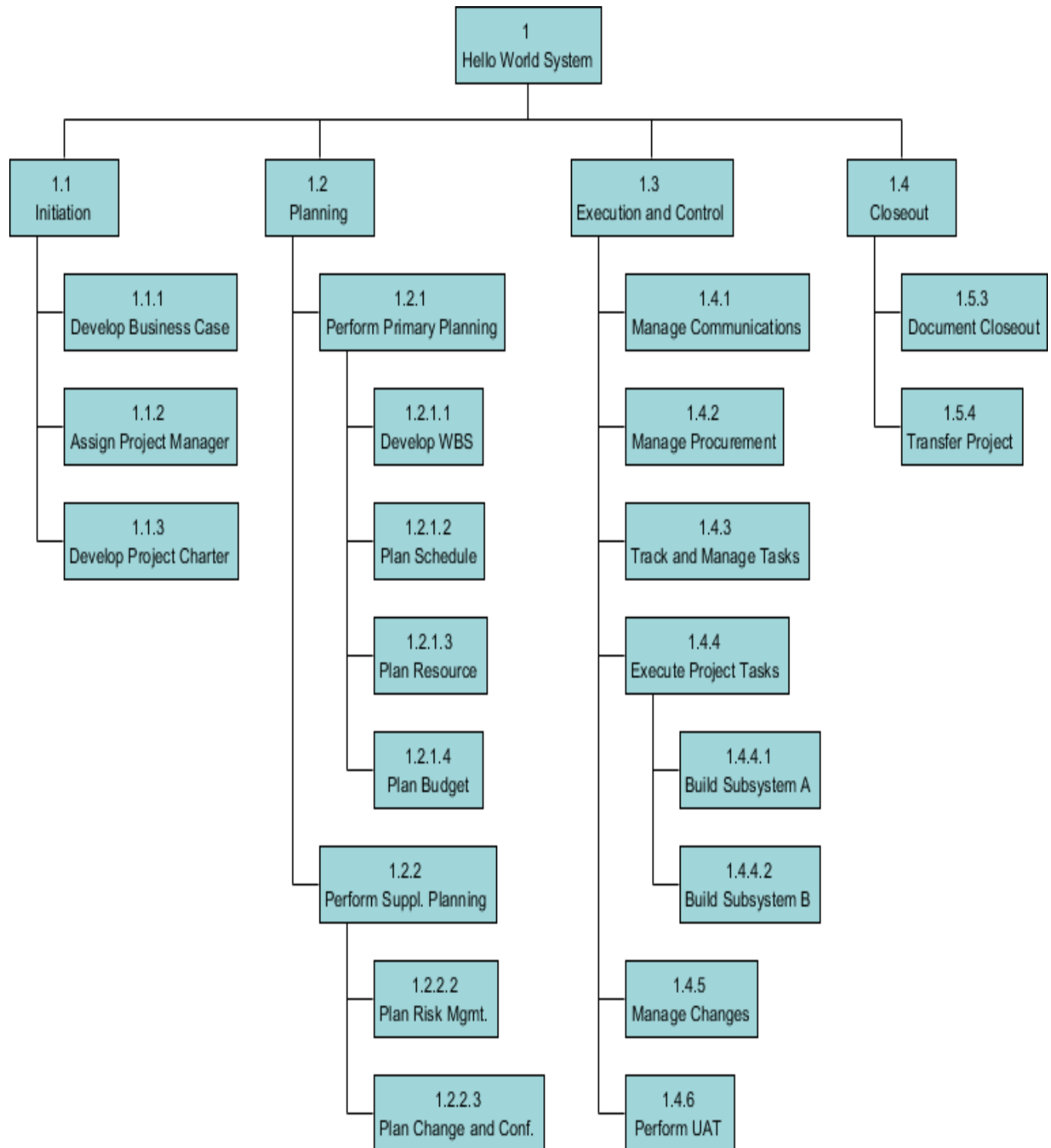
Each **component** within the WBS is referred to as a "**work package.**" A work package is a distinct, measurable unit of work that can be assigned to a specific team or individual. Work packages are at the **lowest level** of the WBS hierarchy.

It helps project managers **identify all the tasks and activities required to complete the project, estimate the resources needed, and create a project schedule.**

A good WBS should exhibit the following characteristics:

- Definable: can be described and easily understood by project participants.
- Manageable: a meaningful unit of work where specific responsibility and authority can be assigned to a responsible individual.
- Estimateable: duration can be estimated in time required to complete, and cost can be estimated in resources required to complete.
- Measurable: can be used to measure progress; has start and completion dates and measurable interim milestones.
- Adaptable: sufficiently flexible so the addition/elimination of work scope can be readily accommodated in the WBS framework.

Work Breakdown Structure example:



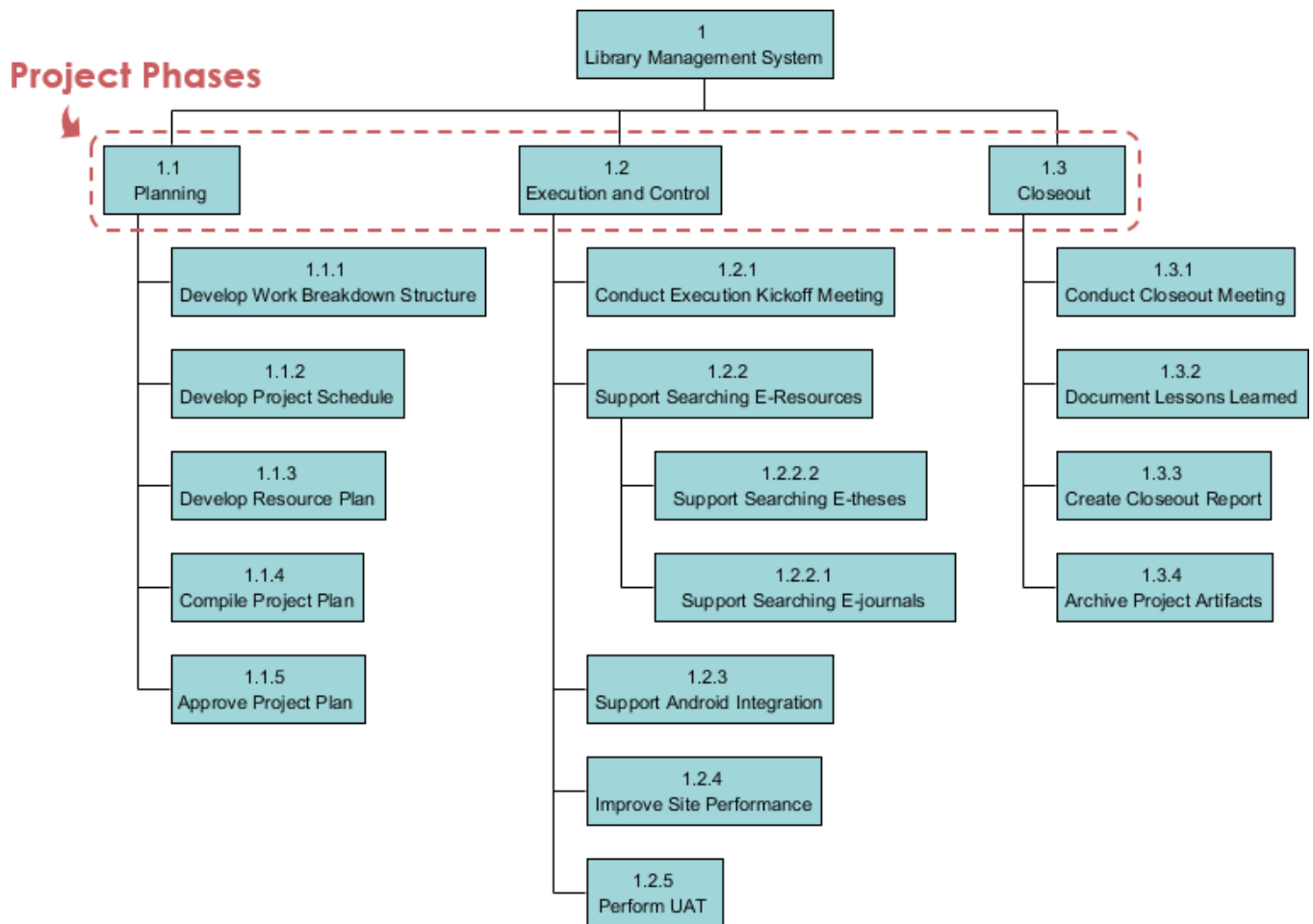
Different Forms of Work Breakdown Structure:

There are **three typical ways in structuring works** with a **Work Breakdown Structure (WBS)**.

They include **phase-based structures, deliverable-based structures and responsibility-based structures**.

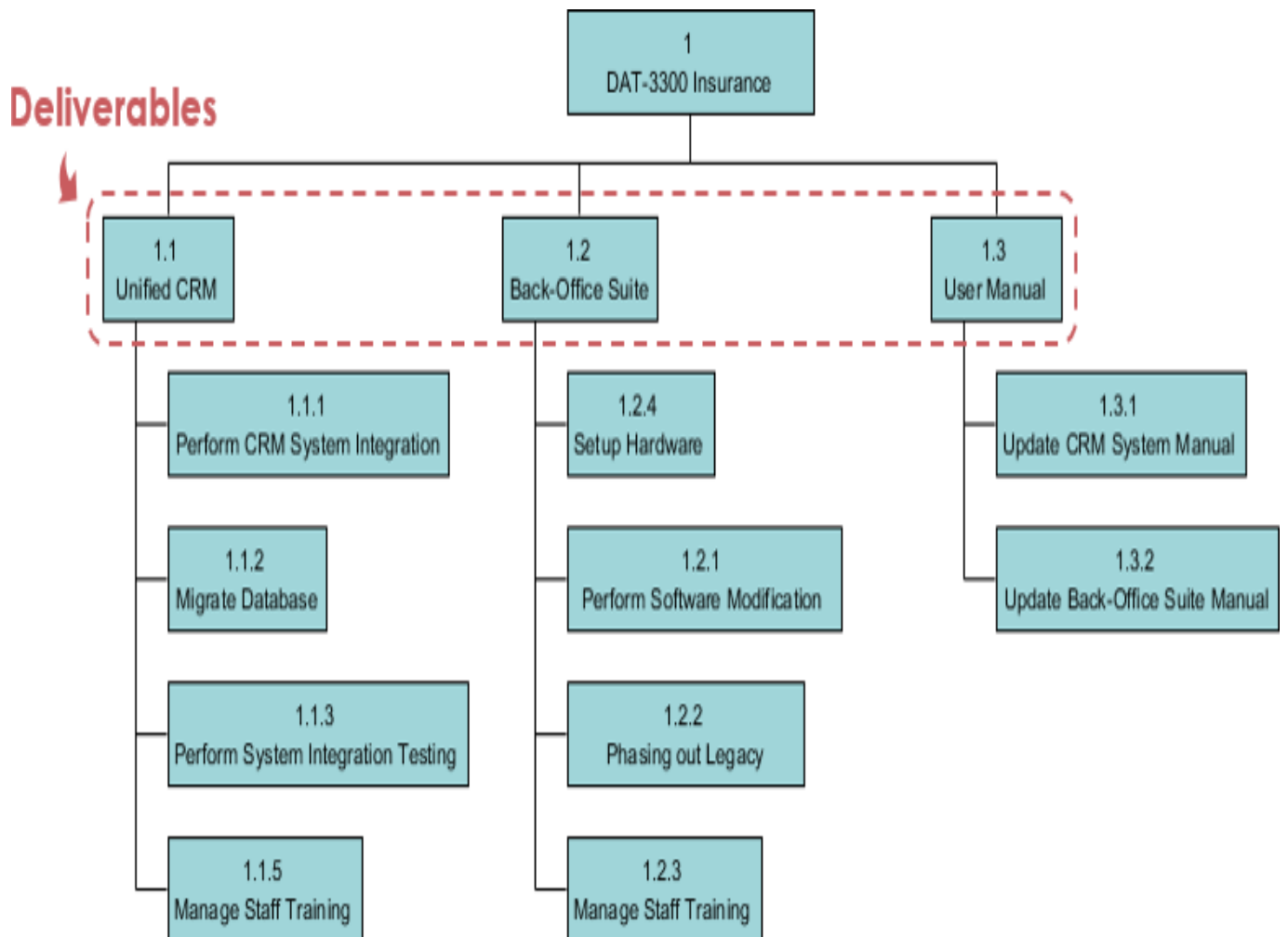
Phase-based structures

Define and structure project activities based on the project phases.



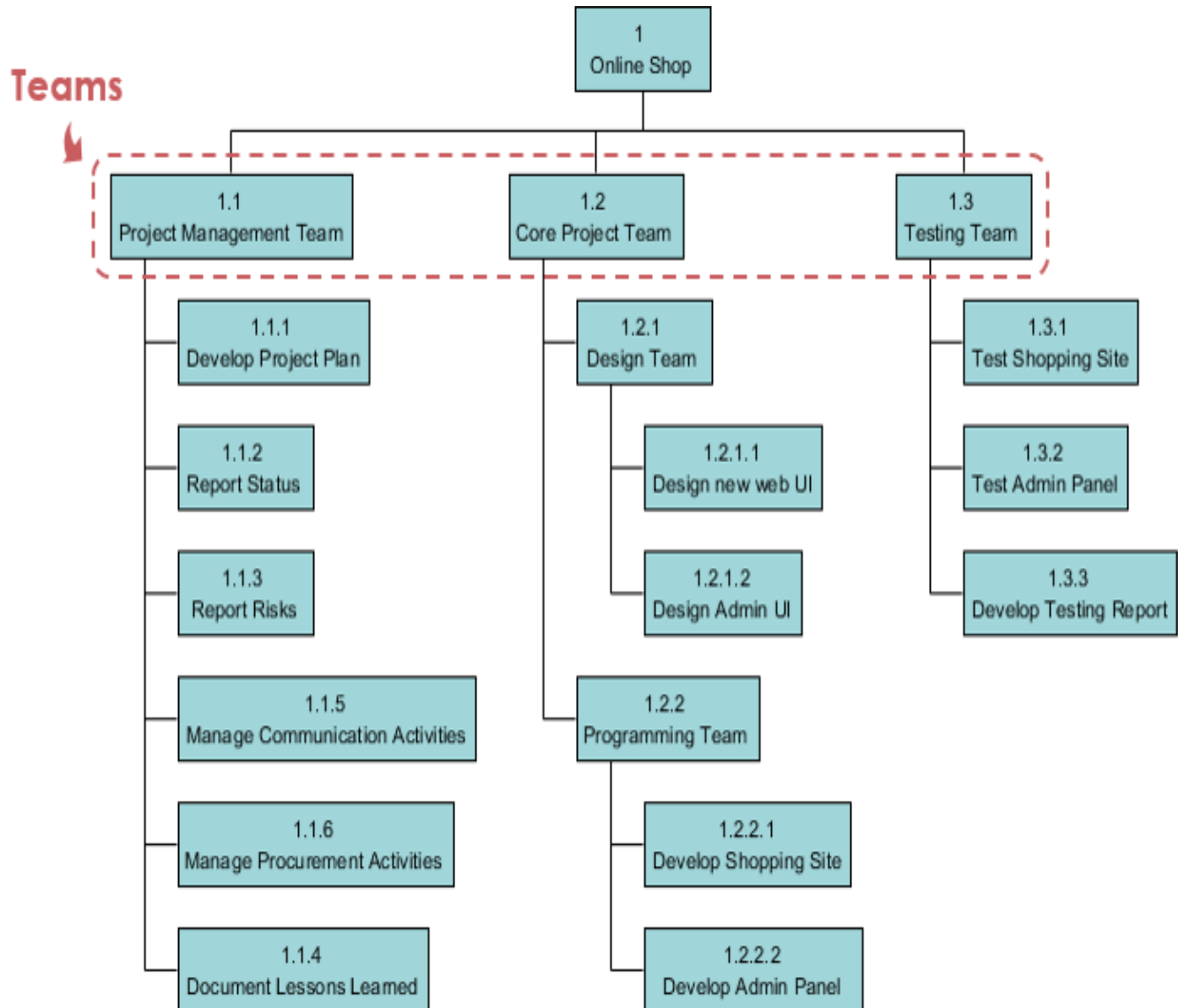
Deliverable-based structures

Define and structure project activities based on the deliverables agreed to deliver.



Responsibility-based structure

Define and structure project activities based on the organization units that will work on the project.



Network Planning Model:

Network planning models are used to represent and manage the dependencies between various tasks or activities involved in a software development project.

These models help project managers create a visual representation of the project's workflow, identify critical paths, allocate resources, and estimate project timelines.

Network planning models are a valuable tool for software project managers, as they can help to ensure that projects are completed on time and within budget.

It is a graphical representation of the project, with nodes representing activities and edges representing the dependencies between them.

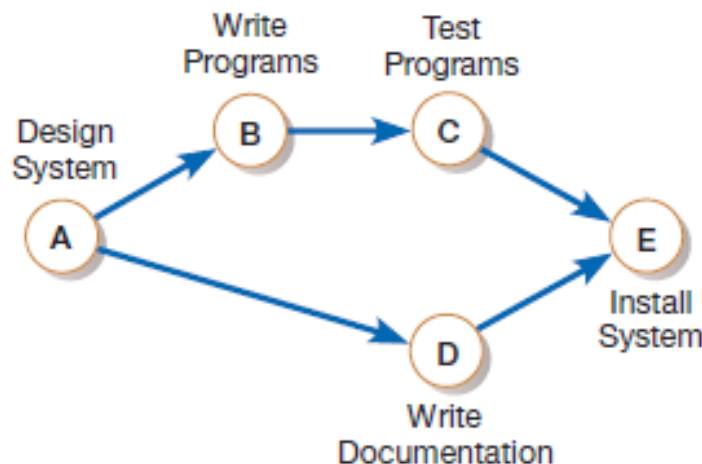


Figure: A network diagram showing activities (represented by circles) and sequence of those activities (represented by arrows)

Network planning models can be used to:

- Identify the critical path of the project.
- Estimate the project duration.
- Develop contingency plans in case of delays.
- Monitor and control the project schedule.

Two widely used network planning models in software project management are the **Critical Path Method (CPM)** and the **Program Evaluation and Review Technique (PERT)**

Critical Path Method (CPM):

- CPM is a **deterministic model** used to **calculate the minimum time** required to complete a project.
- It represents **project activities as nodes** and their **dependencies as directed edges** in a network diagram, often called a "CPM network."
- Each **activity is associated with a duration** and can have dependencies on other activities.
- The **critical path is the longest path** through your network diagram, where the **total duration of activities is the longest**.

- Project managers use CPM to identify critical activities that must be closely monitored and managed to ensure the project finishes on time.

Major steps of the Critical Path Method:

1. Identifying the activities
2. Construct the project network
3. Perform time estimation using forward and backward pass
4. Identify the critical path

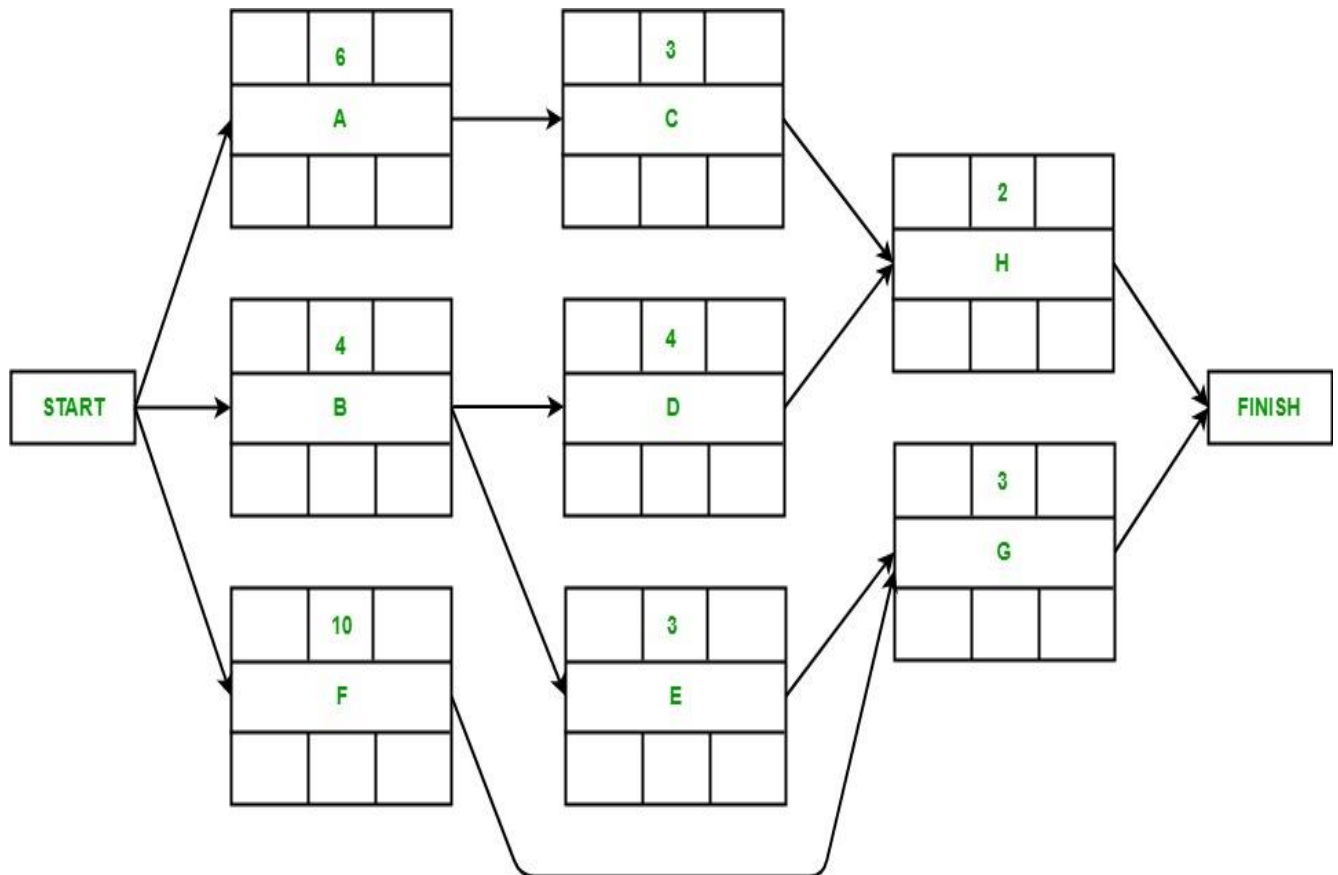
Activity	Duration (in weeks)	Precedents
A	6	–
B	4	–
C	3	A
D	4	B
E	3	B
F	10	–
G	3	E, F
H	2	C, D

Node Representation:

Earliest Start	Duration	Earliest Finish
Activity Label		
Latest Start	Float	Latest Finish

- **Activity label** is the name of the activity represented by that node.
- **Earliest Start** is the date or time at which the activity can be started at the earliest.
- **Earliest Finish** is the date or time at which the activity can be completed at the earliest.
- **Latest Start** is the date or time at which the activity can be started at the latest.
- **Latest Finish** is the date or time at which the activity can be finished at the latest.
- **Float** is equal to the **difference between earliest start and latest start or earliest finish and latest finish**.
- Total **float** represents the **maximum amount of time that an activity within a project can be delayed without causing a delay in the overall project's completion**.

Activity-On-Arrow diagram:



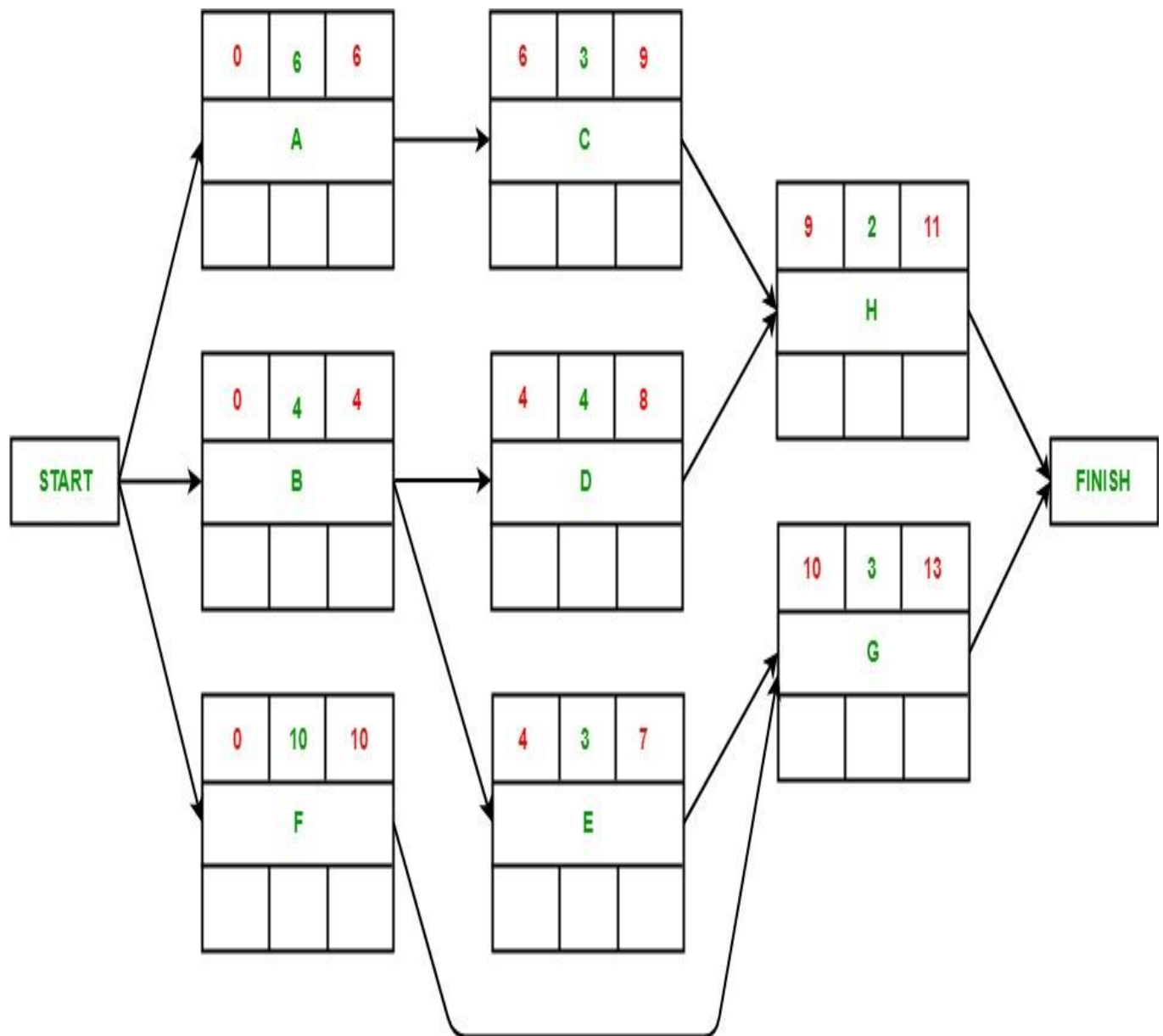
Forward Pass:

The forward pass is carried out to calculate the earliest dates on which each activity may be started and completed.

1. Activity A may start immediately. Hence, earliest date for its start is zero i.e. $ES(A) = 0$. It takes 6 weeks to complete its execution. Hence, earliest it can finish is week 6 i.e. $EF(A) = 6$.

2. Activity B may start immediately. Hence, earliest date for its start is zero i.e. $ES(B) = 0$. It takes 4 weeks to complete its execution. Hence, earliest it can finish is week 4 i.e. $EF(B) = 4$.
3. Activity F may start immediately. Hence, earliest date for its start is zero i.e. $ES(F) = 0$. It takes 10 weeks to complete its execution. Hence, earliest it can finish is week 10 i.e. $EF(F) = 10$.
4. Activity C starts as soon as activity A completes its execution. Hence, earliest week it can start its execution is week 6 i.e. $ES(C) = 6$. It takes 3 weeks to complete its execution. Hence, earliest it can finish is week 9 i.e. $EF(C) = 9$.
5. Activity D starts as soon as activity B completes its execution. Hence, earliest week it can start its execution is week 4 i.e. $ES(D) = 4$. It takes 4 weeks to complete its execution. Hence, earliest it can finish is week 8 i.e. $EF(D) = 8$.
6. Activity E starts as soon as activity B completes its execution. Hence, earliest week it can start its execution is week 4 i.e. $ES(E) = 4$. It takes 3 weeks to complete its execution. Hence, earliest it can finish is week 7 i.e. $EF(E) = 7$.

7. Activity G starts as soon as activity E and activity F completes their execution. Since, activity requires the completion of both for starting its execution, we would consider the $\text{MAX}(\text{ES}(\text{E}), \text{ES}(\text{F}))$. Hence, earliest week it can start its execution is week 10 i.e. $\text{ES}(\text{G}) = 10$. It takes 3 weeks to complete its execution. Hence, earliest it can finish is week 13 i.e. $\text{EF}(\text{G}) = 13$.
8. Activity H starts as soon as activity C and activity D completes their execution. Since, activity requires the completion of both for starting its execution, we would consider the $\text{MAX}(\text{ES}(\text{C}), \text{ES}(\text{D}))$. Hence, earliest week it can start its execution is week 9 i.e. $\text{ES}(\text{H}) = 9$. It takes 2 weeks to complete its execution. Hence, earliest it can finish is week 11 i.e. $\text{EF}(\text{H}) = 11$.



Backward Pass:

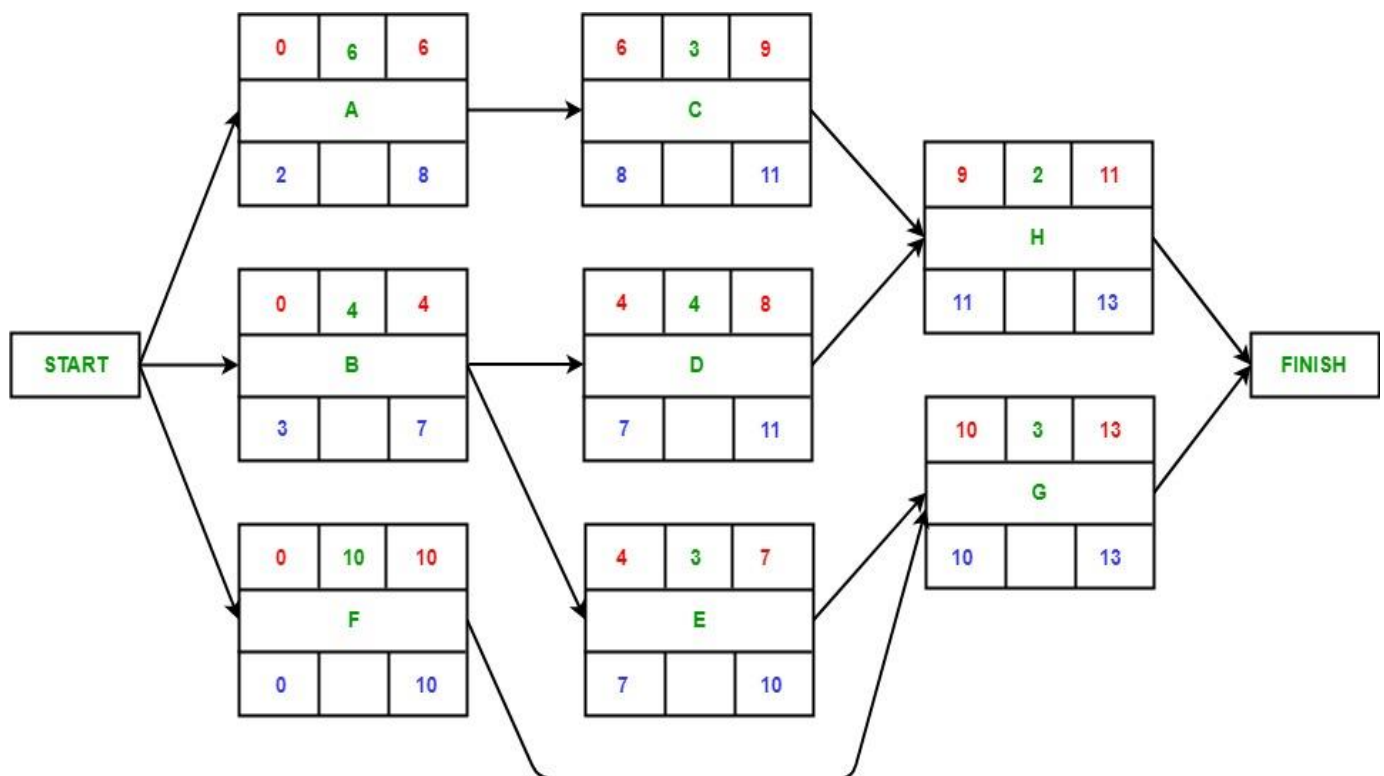
The backward pass is carried out to calculate the **latest dates on which each activity may be started and finished** without delaying the end date of the project. **Assumption: Latest finish date = Earliest Finish date (of project).**

1. Activity G's latest finish date is equal to the earliest finish date of the precedent activity of finish according to the assumption i.e. $LF(G) = 13$. It takes 3 weeks to complete its execution. Hence, latest it can start is week 10 i.e. $LS(G) = 10$.
2. Activity H's latest finish date is equal to the earliest finish date of the precedent activity of finish according to the assumption i.e. $LF(H) = 13$. It takes 2 weeks to complete its execution. Hence, latest it can start is week 11 i.e. $LS(H) = 11$.
3. The latest end date for activity C would be the latest start date of H i.e. $LF(C) = 11$. It takes 3 weeks to complete its execution. Hence, latest it can start is week 8 i.e. $LS(C) = 8$.
4. The latest end date for activity D would be the latest start date of H i.e. $LF(D) = 11$. It takes 4 weeks to complete its execution. Hence, latest it can start is week 7 i.e. $LS(D) = 7$.
5. The latest end date for activity E would be the latest start date of G i.e. $LF(G) = 10$. It takes 3 weeks to complete its execution. Hence, latest it can start is week 7 i.e. $LS(E) = 7$.
6. The latest end date for activity F would be the latest start date of G i.e. $LF(G) = 10$. It takes 10 weeks to

complete its execution. Hence, latest it can start is week 0 i.e. $LS(F) = 0$.

7. The latest end date for activity A would be the latest start date of C i.e. $LF(A) = 8$. It takes 6 weeks to complete its execution. Hence, latest it can start is week 2 i.e. $LS(A) = 2$.

8. The latest end date for activity B would be the earliest of the latest start date of D and E i.e. $LF(B) = 7$. It takes 4 weeks to complete its execution. Hence, latest it can start is week 3 i.e. $LS(B) = 3$.



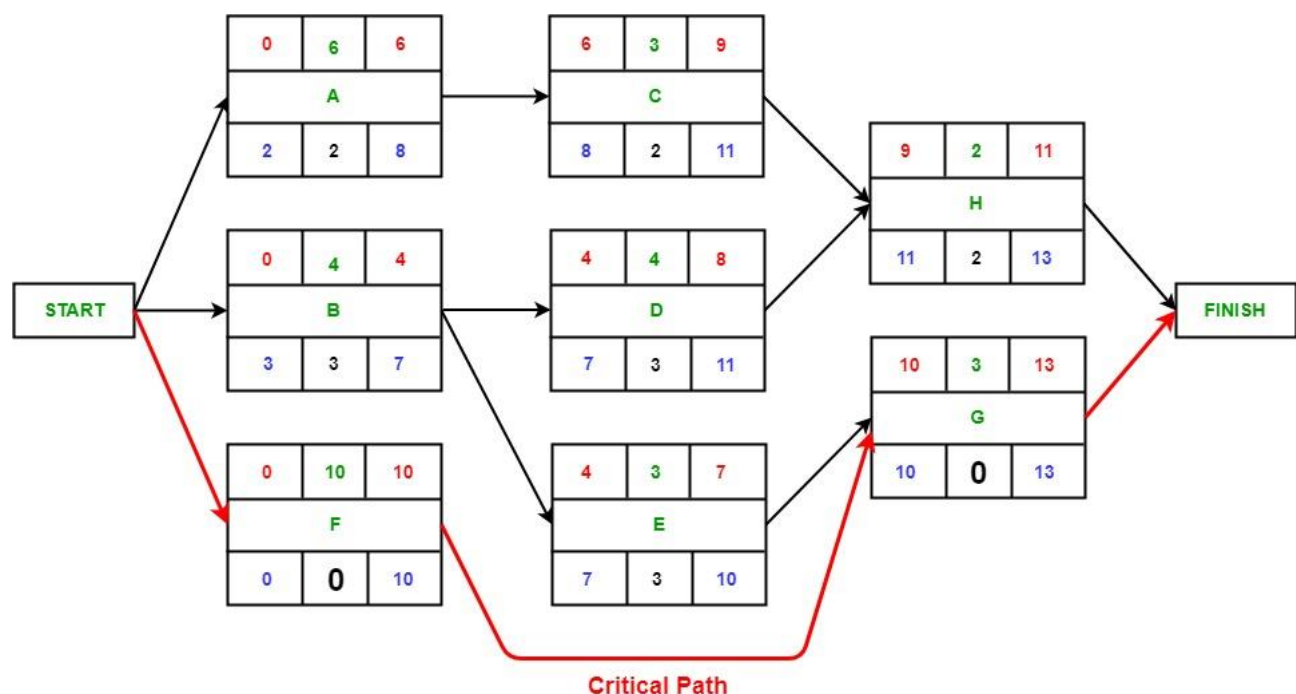
Identifying Critical Path:

Critical path is the path which gives us or helps us to estimate the earliest time in which the whole project can be completed.

Any **delay to an activity on this critical path will lead to a delay in the completion of the whole project.**

In order to identify the critical path, we need to calculate the activity float for each activity.

If the **float of an activity is zero**, then the **activity is an critical activity** and must be added to the critical path of the project network. In this example, activity **F and G have zero float** and hence, are critical activities.



Program Evaluation and Review Technique (PERT):

PERT charts were first created by the U.S. Navy's Special Projects Office in 1950s to guide the Polaris nuclear submarine project.

In project management, the Project Evaluation Review Technique, or PERT, is used to identify the time it takes to finish a particular task or activity.

PERT is a probabilistic method, where the activity times are represented by a probability distribution.

This distribution of activity times is based on three different time estimates made for each activity, which are as follows:

- **Optimistic time:** The shortest possible time to complete the activity.
- **Most likely time:** The most probable time to complete the activity.
- **Pessimistic time:** The longest possible time to complete the activity.

For these three estimates, we have to calculate the expected time of an activity

Expected time $t_e = (t_o + 4t_m + t_p) / 6$

Calculating expected time durations using PERT:

Q.The following table shows the jobs of a network along with their time estimates.

Activity	t_o	t_m	t_p
1-2	6	9	12
1-3	3	4	11
2-4	2	5	14
3-4	4	6	8
3-5	1	1.5	5
2-6	5	6	7
4-6	7	8	15
5-6	1	2	3

- 1.Draw the project network diagram.
- 2.Find the expected duration of each activity.
- 3.Calculate expected project length.

Solution: First of all draw the network diagram for given data as shown below:

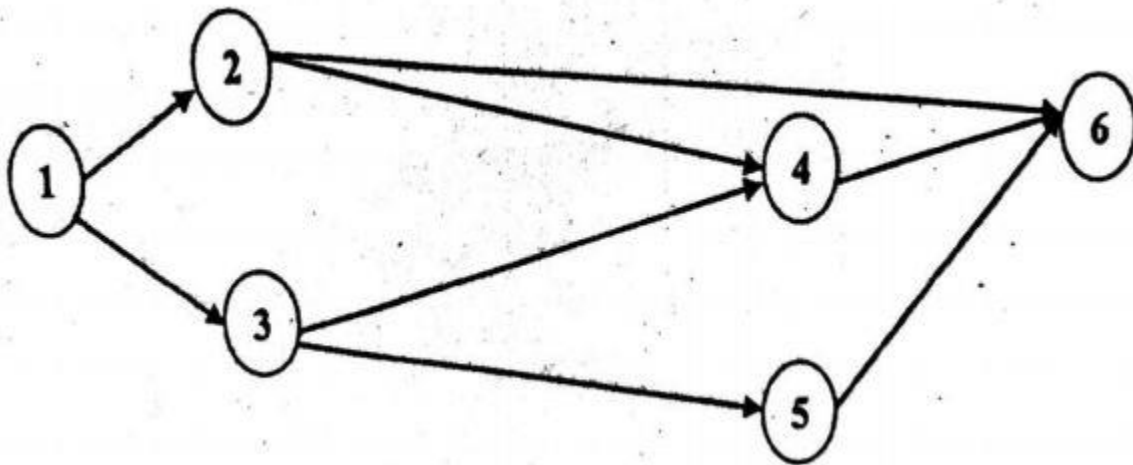


Figure: Network diagram

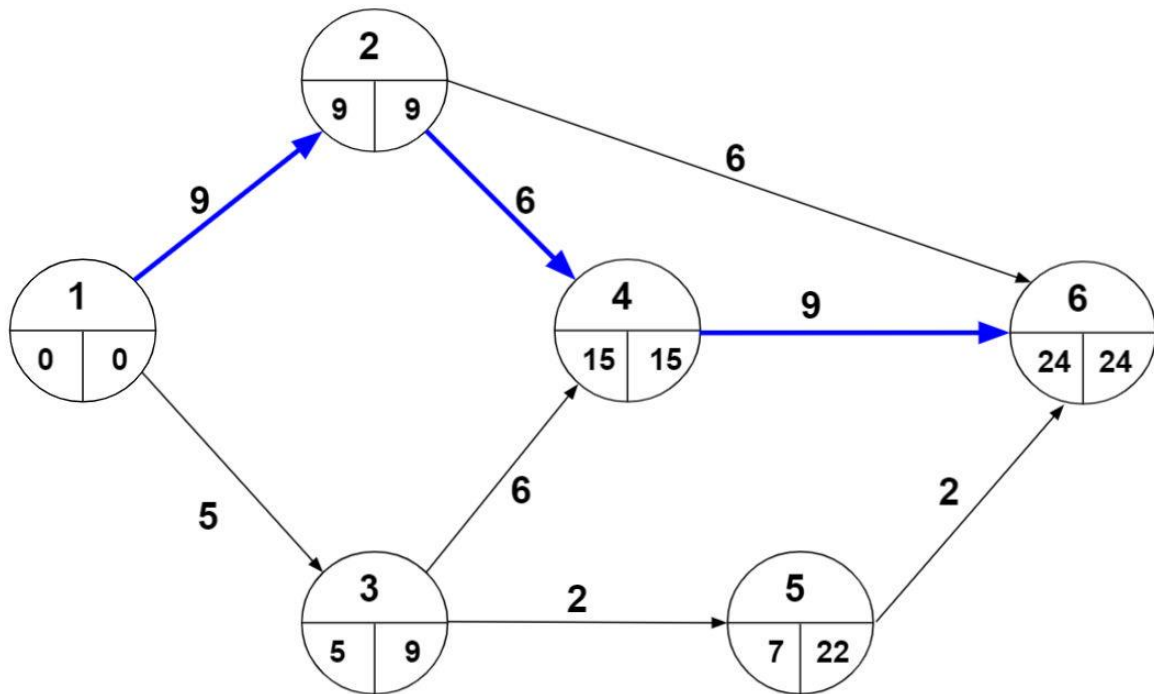
Here the time for completion of activities are probabilistic. So, using given values of time we will find the expected time to completion the activities.

$$\text{Expected time } t_e = (t_o + 4t_m + t_p) / 6$$

For each given activity we will calculate the expected time as follows:

Activity	t_o	t_m	t_p	$t_e = \frac{t_o + 4t_m + t_p}{6}$
1-2	6	9	12	$\frac{6 + 4 \times 9 + 12}{6} = 9$
1-3	3	4	11	5
2-4	2	5	14	6
3-4	4	6	8	6
3-5	1	1.5	5	2
2-6	5	6	7	6
4-6	7	8	15	9
5-6	1	2	3	2

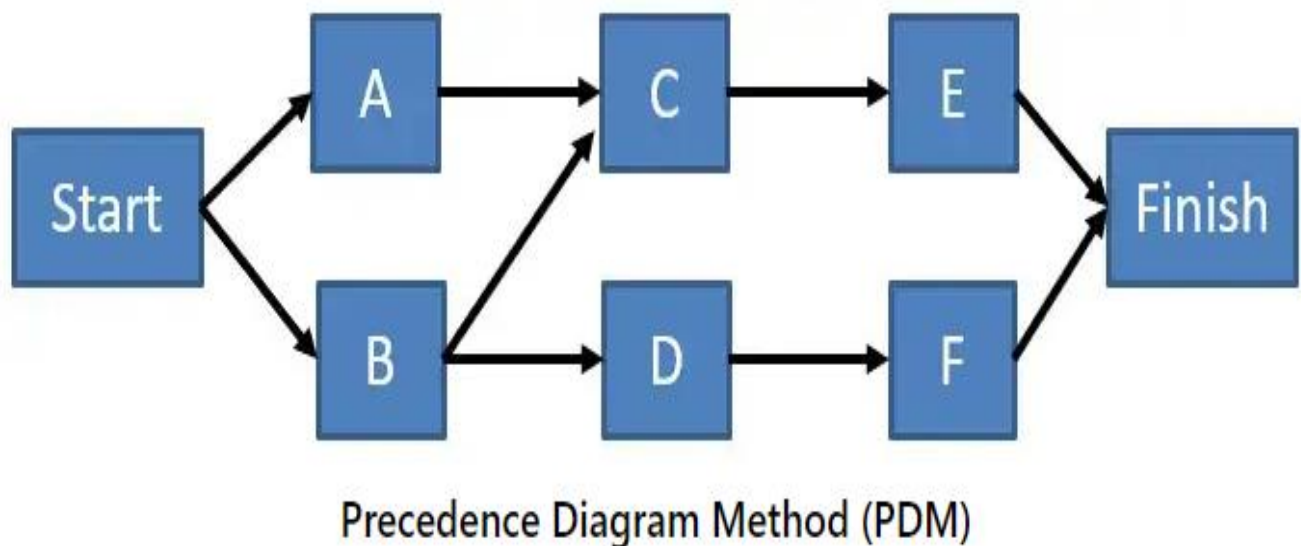
Here the critical path is along the activities 1-2, 2-4, 4-6. So the critical path is 1-2-4-6. Following diagram is prepared to show critical path along with EST and LFT.



Expected project length is 24 days.

Precedence Diagramming Method (PDM):

A strategy for scheduling activities in a project plan, the precedence diagramming method (PDM) is a strategy for developing a project schedule network diagram that utilizes nodes to represent activities and associates them with projectiles that illustrate the dependencies. This method is the activity-on-node (AON).



Precedence Diagram Method (PDM) Benefits

There are a lot of benefits that can be obtained by using the PDM. These are:

- Highlights relationships and dependencies among activities to ensure planning efficiency.

- Identifies possible missing activities.
- Helps identify critical activities to ensure better planning.
- Helps develop the overall project schedule.
- Good communication tool for project team members.

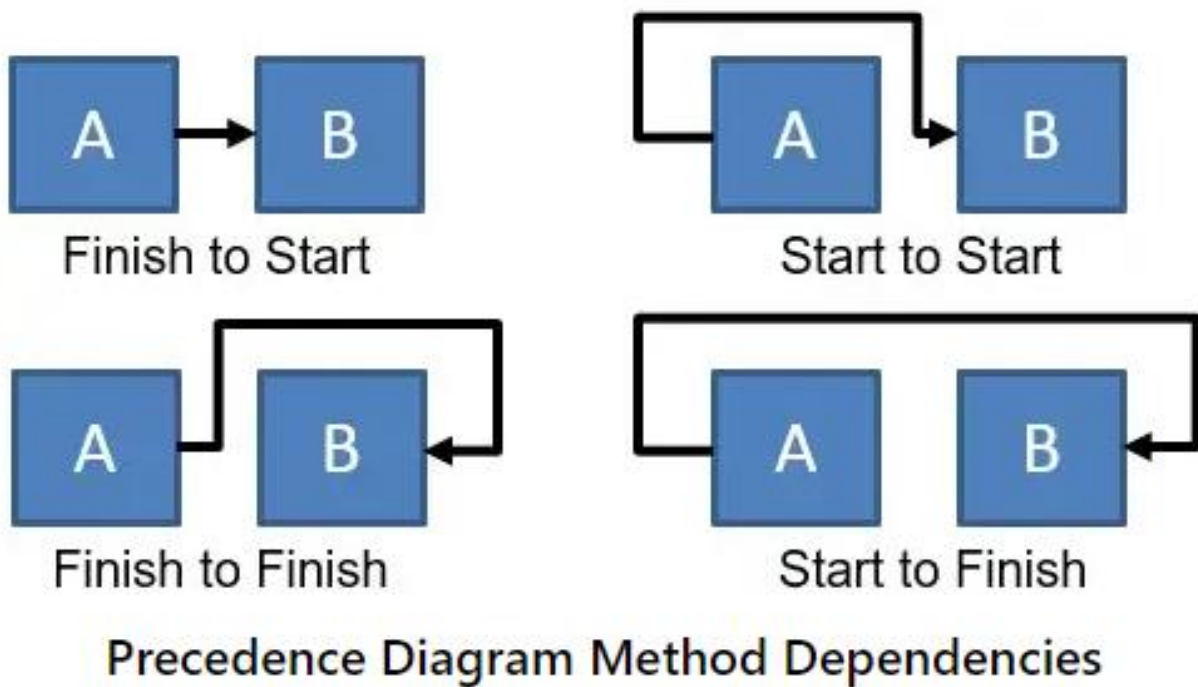
The PDM has four ways of developing the diagram.
These methods are:

Finish-Start: In this dependency, an activity cannot start before a previous activity has ended. This is the most commonly used dependency.

Start-Start: In this dependency, there is a defined relationship between the start of activities.

Finish-Finish: In this dependency, there is a defined relationship between the end dates of activities.

Start-Finish: In this dependency, there is a defined relationship between the start of one activity and the end date of a successor activity. This dependency is rarely used.



Steps to Developing a Precedence Diagram

- **Step 1:** Break your Work Breakdown Structure (WBS) into activity levels.
- **Step 2:** List all activities and their sequences in a table.
- **Step 3:** Add relationships and dependencies to each activity in the table.
- **Step 4:** Draw the diagram.